Question ID: bbb77c84

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Words in Context	Hard

Question

Containing over 160 billion base pairs of DNA resulting from polyploidy (whole genome duplication) and accumulation of noncoding DNA without significant deletion, the genome of *Tmesipteris oblanceolata* (New Caledonian fork fern) is the largest known among eukaryotes—and, given the high biological costs of extremely large genomes, is unlikely to be ______.

Which choice completes the text with the most logical and precise word or phrase?

Answer

A. protracted

B. superseded

C. curtailed

D. obviated

Correct Answer: B

Rationale

Choice B is the best answer because it most logically completes the text's discussion of the fork fern's enormous genome. In this context, "superseded" means surpassed or exceeded by something else. The text establishes that this plant has "the largest known" genome among eukaryotes and mentions "the high biological costs of extremely large genomes," suggesting there are natural constraints that make it biologically disadvantageous to maintain such massive genetic material. This context supports the idea that it's unlikely another organism will evolve an even larger genome that would replace this plant's record—that is, its status as having the largest genome is unlikely to be superseded.

Choice A is incorrect because in this context, "protracted" would mean drawn out over time or space, and the text focuses on the record size of the New Caledonian fork fern's genome and the fact that extremely large genomes carry "high biological costs." This information suggests that the biological upper limit for genome size may have been reached and that the New Caledonian fork fern's genome is therefore unlikely to be surpassed, or superseded, not that it's unlikely to be protracted. Choice C is incorrect because in this context, "curtailed" would mean cut short or reduced, and the text focuses on the record size of the New Caledonian fork fern's genome and the fact that extremely large genomes carry "high biological costs." This information suggests that the biological upper limit for genome size may have been reached and that the New Caledonian fork fern's genome is therefore unlikely to be surpassed, or superseded, not that it's unlikely to be curtailed. Choice D is incorrect because in this context, "obviated" would mean made unnecessary or redundant, and the text focuses on the record size of the New Caledonian fork fern's genome and the fact that extremely large genomes carry "high biological costs." This information suggests that the biological upper limit for genome size may have been reached and that the New Caledonian fork fern's genome is therefore unlikely to be surpassed, or superseded, not that it's unlikely to be obviated.